

Name: Savitha R

Reg.no: 192421368

13) Create an Interactive Scientific Calculator HTML buttons and input fields, CSS Grid or Flexbox, JavaScript functions, arithmetic operators, Math object and click events.

Program:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Scientific Calculator</title>

<style>
body{
    margin:0;
    font-family:Arial,sans-serif;
    background:linear-gradient(to right,#36D1DC,#5B86E5);
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
}

.calculator{
    background:white;
    padding:20px;
    border-radius:15px;
    box-shadow:0 0 20px gray;
```

```
width:350px;  
}
```

```
h2{  
  text-align:center;  
  color:#1565c0;  
}
```

```
#display{  
  width:100%;  
  height:50px;  
  font-size:22px;  
  text-align:right;  
  margin-bottom:15px;  
  border:2px solid #1565c0;  
  border-radius:5px;  
  padding-right:10px;  
}
```

```
.buttons{  
  display:grid;  
  grid-template-columns:repeat(4,1fr);  
  gap:10px;  
}
```

```
button{  
  padding:15px;  
  font-size:18px;  
  border:none;  
  border-radius:8px;  
  cursor:pointer;
```

```
background:#1976d2;
color:white;
}
```

```
button:hover{
background:#0d47a1;
}
```

```
.equal{
background:green;
}
```

```
.clear{
background:red;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="calculator">
```

```
<h2>Scientific Calculator</h2>
```

```
<input type="text" id="display" readonly>
```

```
<div class="buttons">
```

```
<button onclick="addValue('7')">7</button>
```

```
<button onclick="addValue('8')">8</button>
```

<button onclick="addValue('9')">9</button>

<button onclick="addValue('/')">÷</button>

<button onclick="addValue('4')">4</button>

<button onclick="addValue('5')">5</button>

<button onclick="addValue('6')">6</button>

<button onclick="addValue('*')">×</button>

<button onclick="addValue('1')">1</button>

<button onclick="addValue('2')">2</button>

<button onclick="addValue('3')">3</button>

<button onclick="addValue('-')">-</button>

<button onclick="addValue('0')">0</button>

<button onclick="addValue('.')">.</button>

<button class="equal" onclick="calculate()">=</button>

<button onclick="addValue('+')">+</button>

<button onclick="squareRoot()">√</button>

<button onclick="power()">x²</button>

<button onclick="sinValue()">sin</button>

<button onclick="cosValue()">cos</button>

<button onclick="tanValue()">tan</button>

<button onclick="logValue()">log</button>

<button onclick="clearDisplay()" class="clear">C</button>

<button onclick="backspace()">⌫</button>

</div>

</div>

```
<script>
```

```
function addValue(value){  
  document.getElementById("display").value += value;  
}
```

```
function calculate(){  
  try{  
    document.getElementById("display").value=  
    eval(document.getElementById("display").value);  
  }  
  catch{  
    alert("Invalid Expression");  
  }  
}
```

```
function clearDisplay(){  
  document.getElementById("display").value="";  
}
```

```
function backspace(){  
  let d=document.getElementById("display");  
  d.value=d.value.slice(0,-1);  
}
```

```
function squareRoot(){  
  let d=document.getElementById("display");  
  d.value=Math.sqrt(Number(d.value));  
}
```

```
function power(){  
let d=document.getElementById("display");  
d.value=Math.pow(Number(d.value),2);  
}
```

```
function sinValue(){  
let d=document.getElementById("display");  
d.value=Math.sin(Number(d.value)*Math.PI/180).toFixed(4);  
}
```

```
function cosValue(){  
let d=document.getElementById("display");  
d.value=Math.cos(Number(d.value)*Math.PI/180).toFixed(4);  
}
```

```
function tanValue(){  
let d=document.getElementById("display");  
d.value=Math.tan(Number(d.value)*Math.PI/180).toFixed(4);  
}
```

```
function logValue(){  
let d=document.getElementById("display");  
d.value=Math.log10(Number(d.value)).toFixed(4);  
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:

